Energy flow and power exchange

1 Problem Setup

The exterior calculation supplies an in-plane electric field $\boldsymbol{E} = (E_x, E_y, 0)$ throughout the mesh. The magnetic calculation supplies $\boldsymbol{B} = (0, 0, B_z)$. We want to show where electromagnetic energy flows and where the battery supplies energy that the conductor absorbs as heat.

2 Deriving the energy-flow vector

In this model $\boldsymbol{H} = \boldsymbol{B}/\mu$. The Poynting vector is the cross product

$$\boldsymbol{S} = \boldsymbol{E} \times \boldsymbol{H} = \frac{B_z}{\mu}(E_y, -E_x, 0). \tag{1}$$

The electric field is constant on each triangle, while B_z is linear there. Thus $\boldsymbol{S}$ is linear within a triangle. Its exact triangle average is

$$\overline{\boldsymbol{S}}_h = \frac{\overline{B}_{z,h}}{\mu}(E_{y,h}, -E_{x,h}), \qquad \overline{B}_{z,h} = \frac{b_{h,0} + b_{h,1} + b_{h,2}}{3}. \tag{2}$$

The app draws arrows from these averages. They describe energy flow, not the motion of electrons. In the strictly two-dimensional model, $B_z = 0$ in the outside air, so $\boldsymbol{S} = 0$ there even though $\boldsymbol{E}$ may be nonzero. The cavity can have nonzero energy flow because its magnetic field is generally nonzero.

To connect this flow to the circuit, use the vector identity

$$\nabla \cdot (\boldsymbol{E} \times \boldsymbol{H}) = \boldsymbol{H} \cdot (\nabla \times \boldsymbol{E}) - \boldsymbol{E} \cdot (\nabla \times \boldsymbol{H}). \tag{3}$$

In steady state $\nabla \times \boldsymbol{E} = 0$, and Ampere's law gives $\nabla \times \boldsymbol{H} = \boldsymbol{J}$. Therefore

$$\nabla \cdot \boldsymbol{S} = -\boldsymbol{J} \cdot \boldsymbol{E}. \tag{4}$$

The relation says that energy leaving a small volume through the field is balanced by electrical work inside it.

3 Power supplied by the battery and absorbed as heat

Inside the conductor, the current law is $\rho\boldsymbol{J} = \boldsymbol{E} + \boldsymbol{F}$. Rearranging and taking the dot product with $\boldsymbol{J}$ gives

$$\boldsymbol{J} \cdot \boldsymbol{E} = \rho|\boldsymbol{J}|^2 - \boldsymbol{J} \cdot \boldsymbol{F}. \tag{5}$$

The term $\rho|\boldsymbol{J}|^2$ is Joule heating in the wire, battery, and resistors. The term $\boldsymbol{J}\cdot\boldsymbol{F}$ is power supplied by the battery's driving field and is zero outside the battery. The app's power-exchange view shows

$$p_{\text{exchange,h}} = \boldsymbol{J}_h \cdot \boldsymbol{F}_h - \rho_h|\boldsymbol{J}_h|^2. \tag{6}$$

Positive values mean net supply, while negative values mean net absorption as heat. In air, current and both power-density terms are zero.

The battery-power and resistor-heating readouts sum the corresponding triangle densities times area. Let $\mathcal{T}_B$ be the set of battery triangles and $\mathcal{T}_R$ the set of resistor triangles:

$$P'_{\text{battery}} = \sum_{h\in\mathcal{T}_B} A_h\, \boldsymbol{J}_h \cdot \boldsymbol{F}_h, \qquad P'_{\text{resistors}} = \sum_{h\in\mathcal{T}_R} A_h\, \rho_h|\boldsymbol{J}_h|^2. \tag{7}$$

These powers are per unit depth in the unchanged z direction. Since EMF is work per unit charge, the displayed loop current is $I' = P'_{\text{battery}}/\mathcal{E}$ when $\mathcal{E} \neq 0$.

Code: `calculations/poynting.py`.